Mapped Files, Cleaning Policy, and Virtual Memory Interface

1 Core Idea

Three Related Design Issues

This note covers three virtual-memory design issues:

- **memory-mapped files**: files accessed through virtual memory;
- **cleaning policy**: keeping free clean frames available;
- **virtual memory interface**: giving programmers controlled access to memory mappings.

2 Memory-Mapped Files

Definition

A **memory-mapped file** is a file mapped into a process's virtual address space so the process can access the file as if it were memory.

Instead of calling `read` and `write`, the program can treat the file like a large array of bytes.

Main Idea

$\boxed{\text{file on disk} \longleftrightarrow \text{region of virtual address space}}$

3 How Mapping Works

The process asks the OS to map a file into part of its virtual address space.

1. The process issues a system call to map the file.
2. The OS creates virtual-memory mappings for the file region.
3. Usually, no file pages are loaded immediately.
4. When the process touches a mapped page, a page fault occurs.
5. The OS demand-pages that part of the file into RAM.
6. Modified mapped pages are written back to the file when unmapped or when the process exits.

Demand Paging

Mapped files are usually loaded page by page, only when touched.

4 Mapped Files as an I/O Model

Mapped files provide an alternative to explicit file I/O.

I/O model	How the program accesses data
Traditional I/O	Calls `read` and `write`.
Memory-mapped I/O	Reads and writes memory addresses in the mapped region.

Convenience

For some programs, treating a file as a byte array in memory is simpler than manually issuing reads and writes.

5 Sharing with Mapped Files

If two processes map the same file, they can share memory through it.

Action	Effect
Process A writes to mapped region	The shared memory changes.
Process B reads same mapped region	It sees the data written by A.
Both map a scratch file	The file acts as a shared-memory communication channel.

High-Bandwidth Communication

Mapped files can provide fast interprocess communication because data can be shared through memory instead of copied through pipes or messages.

6 Shared Libraries as Mapped Files

Shared libraries are a special case of mapped files.

Connection

A shared library can be mapped into the virtual address spaces of multiple processes. The library file on disk acts as the backing store.

Only the library pages that are actually used need to be brought into RAM.

7 Cleaning Policy

Definition

A **cleaning policy** keeps enough free, clean page frames available so page faults can be handled quickly.

Paging works best when the OS has free frames ready. If every frame is full and dirty, a page fault may need to wait for a disk write before the needed page can be loaded.

8 Paging Daemon

Paging Daemon

A **paging daemon** is a background process that periodically checks memory and prepares free page frames.

The paging daemon usually sleeps. When too few frames are free, it wakes up and starts selecting pages to evict.

Page state	Daemon action
Clean page	Can be moved to the free-frame pool quickly.
Dirty page	Must be written back to disk before becoming a clean reusable frame.

Why This Helps

It is faster to have clean free frames ready before a page fault than to start cleaning only after memory is urgently needed.

9 Remembering Evicted Pages

When a page is evicted, the OS may remember what was in that frame.

Reclaiming

If an evicted page is needed again before its frame is overwritten, the OS can reclaim it from the free-frame pool instead of reading it again from disk.

This improves performance because some evicted pages may be referenced again soon.

10 Two-Handed Clock Cleaning

One cleaning method uses a two-handed clock.

Clock hand	Role
Front hand	Controlled by the paging daemon. It writes dirty pages back to disk or skips clean pages.
Back hand	Used for page replacement. It is more likely to find clean pages because the front hand has already cleaned ahead.

Two-Handed Clock Idea

front hand cleans pages $\rightarrow$ back hand finds clean victims more easily

11 Virtual Memory Interface

Definition

A **virtual memory interface** gives programmers some control over the memory map instead of making virtual memory completely transparent.

Most of the time, programs simply see a large virtual address space. In advanced systems, programs can explicitly map, unmap, share, or transfer memory regions.

12 Controlled Shared Memory

If processes can name memory regions, one process can give another process access to the same pages.

Operation	Result
Process A names a memory region	The region can be identified by the OS.
Process A gives the name to process B	B can map the same region.
Both access the region	They communicate through shared memory.

13 Message Passing by Page Remapping

Normally, message passing copies data from one address space to another. Page remapping can avoid copying the actual data.

1. Sender places the message in one or more pages.
2. Sender unmaps those pages.
3. Receiver maps those pages into its own address space.
4. Only page mapping information is transferred, not the full message bytes.

Performance Benefit

For large messages, remapping pages can be much faster than copying all the data.

14 Distributed Shared Memory

Definition

Distributed shared memory lets processes on different machines share pages over a network.

The system tries to make remote pages appear like shared memory.

1. A process references a page not currently mapped locally.
2. A page fault occurs.
3. The page fault handler locates the machine holding the page.
4. That machine unmaps or transfers ownership of the page.
5. The page is sent over the network.
6. The receiving machine maps it in.
7. The faulting instruction restarts.

Cost

Distributed shared memory hides communication behind page faults, but network transfer is much slower than local memory access.

15 Final Summary

One-Box Summary

Mapped files map disk files into virtual memory.
They support convenient I/O and shared-memory communication.
Cleaning policy keeps clean free frames ready for page faults.
A paging daemon can clean pages in the background.
Advanced VM interfaces let programs share or remap pages directly.